CURRICULUM VITAE

Mohammad M. T.R. Hussein

Biomedical Engineering & Bioinformatics | Computational Imaging | Digital Holographic Microscopy

KAUST, Saudi Arabia
mohammad.mtr.hussein@gmail.com
github.com/MoeHussein
ORCID 0009-0009-7211-0974

PROFESSIONAL SUMMARY

Biomedical engineer with an MSc in Biomedical Engineering and Bioinformatics, currently a Visiting Student Researcher at KAUST. Focused on optical and computational imaging, with experience in optical system development, polarization-sensitive digital holographic microscopy, Python-based image and data processing, full-field Jones matrix reconstruction, and current work involving iSCAT and TIRF microscopy systems.

RESEARCH EXPERIENCE

15 OCT 2026 - 15 APR 2027

Visiting Student Researcher (VSRP) - Optical Microscopy for Biomedical Applications

King Abdullah University of Science and Technology (KAUST)

Development of Label-Free Optical Microscopy for Biomedical Applications | Prof. Wen Wei Wayne Yang

- Contributing to the development of optical microscopy systems for biomedical applications within the KAUST Visiting Student Research Program.
- Current work includes, but is not limited to, interferometric scattering microscopy (iSCAT) and total internal reflection fluorescence (TIRF) microscopy systems.
- The broader project targets quantitative data acquisition, computational analysis, automated feature extraction, and validation using biological or clinical samples.

15 SEP 2023 - SEP 2026

Graduate Researcher - Biomedical Optics & Computational Imaging

Istanbul Medipol University

TÜBİTAK 2530 Türkiye-Moldova Bilateral Cooperation, Project 123N774

- Developed a polarization-sensitive DHM system and Python pipeline for FFT hologram reconstruction, phase processing, full-field Jones matrix reconstruction, retardance and fast-axis analysis.
- Led the end-to-end experimental workflow in collaboration with my academic advisor, including system design and setup, equipment planning, camera configuration, experiments, image processing, validation, visualization and manuscript preparation.
- Published a dual-shot phase-bias equalization method validated with birefringent film and applied to mouse brain and human cornea samples.

SELECTED PUBLICATION

PEER-REVIEWED ARTICLE | Q1 IN THE SJR OPTICS CATEGORY

Dual-shot polarization digital holographic microscopy with phase bias equalization for full-field Jones matrix imaging

Mohammad M. T. R. Hussein and M. Fatih Toy | Biomedical Optics Express 17, 1877-1896 | 2026

DOI: doi.org/10.1364/BOE.585564

TECHNICAL SKILLS

Optical & computational imaging

PDHM, iSCAT microscopy, TIRF microscopy, Jones matrix reconstruction, Polarization analysis, FFT hologram reconstruction, Phase processing, Retardance and optical-axis analysis

Programming

Python, Scientific image processing, Numerical analysis, Scientific visualization

Medical 3D & simulation

DICOM, 3D Slicer, Fusion 360, ANSYS static structural, Finite element analysis, 3D printing

Design & product

Web design, Brand identity, Product prototyping, Research communication

LANGUAGES

- **English**: Academic level
- **Arabic**: Native
- **Turkish**: Upper-intermediate
- **Japanese**: Beginner

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SELECTED ENGINEERING & PRODUCT PROJECTS

MANUSCRIPT IN PREPARATION

Ongoing Computational Imaging Research

Optimization-based methods for quantitative imaging.

The core method and manuscript draft are complete. Technical details are withheld pending submission.

Optimization, Python, Computational imaging

UNDERGRADUATE COMPUTATIONAL DESIGN AND SIMULATION

Patient-Specific Knee Reconstruction & FEA

Patient-specific knee models, an implant concept and comparative structural simulations.

Reconstructed patient-specific knee models, proposed a new implant concept and completed FEA comparisons covering deformation, displacement and von Mises stress. The work was computational and not clinically validated.

DICOM, 3D Slicer, Fusion 360, ANSYS, FEA, 3D printing

PAUSED STARTUP

VitroMech

An early biomedical 3D modeling concept focused on patient-specific anatomy.

Built and presented an early startup concept. The startup is currently paused and no clinical outcomes are claimed.

Medical imaging, DICOM, 3D modeling, Product design, Web design

PERSONAL PRODUCT

Workin

An evidence-informed, phone-first calisthenics workout tracker.

Delivered a public tracker with day and week progress, rest timers, notes and downloadable plans.

Product design, Next.js, TypeScript, Local-first data, GitHub Pages

PERSONAL PRODUCT

Salat Plans

A task planner that organizes daily priorities around prayer times.

Released a working local-first planner with prayer-based time blocks and a focused daily planning experience.

Product design, React, TypeScript, Vite, Supabase, Vercel

EDUCATION

15 SEP 2023 - SEP 2026

MSc | Biomedical Engineering and Bioinformatics

Istanbul Medipol University

- Awarded Sep 2026
- Thesis focus: polarization-sensitive digital holographic microscopy

11 SEP 2019 - 10 JUN 2023

BSc | Biomedical Engineering

Istanbul Medipol University

- CGPA 3.35/4.00
- Graduated with Honors
- Ranked 3rd in my class

RECOGNITION & PRESENTATIONS

12 SEP 2025

Third Place | Best Poster Competition | FOTONİK 2025

Enhancing Jones Matrix Reconstruction in Polarization DHM via Phase Bias Equalization | Koç University.

27-29 AUG 2025

Oral presentation | NanoTR-19

Jones Matrix Analysis via a Polarization-Sensitive Sensor in Digital Holographic Microscopy.

ONGOING INDEPENDENT LEARNING

- Stanford CS231N | Deep Learning for Computer Vision
- OSSU Computer Science
- The Complete Full-Stack Web Development Bootcamp
- KIT Computational Imaging